

Finite lattices and large inductive dimension

D. N. GEORGIOU^a, Y. HATTORI^b, A. C. MEGARITIS^a AND F. SERETI^a

^a University of Patras, Department of Mathematics, 26504 Patra, Greece

(georgiou@math.upatras.gr, acmegaritis@upatras.gr, seretifot@gmail.com)

^b Shimane University (Professor Emeritus), Matsue, 690-8504, Japan (hyasu2014@gmail.com)

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ABSTRACT

The Ordered Set Theory is a branch of Mathematics that studies partially ordered sets (usually posets) and lattices. The meaning of dimension is one of the main parts of this field. In particular, the covering dimension, the Krull dimension and the small inductive dimension have been studied extensively for the class of finite lattices. In this paper, we insert a new meaning of dimension for finite lattices called large inductive dimension. We study various of its properties based on minimal covers. Also, given two finite lattices, we study the dimension Ind of their linear sum, Cartesian, lexicographic and rectangular product, investigating the “behavior” of this dimension. In addition, we study relations of this new dimension with the small inductive dimension, covering dimension and Krull dimension, presenting various facts and examples that strengthen the corresponding results.

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1. INTRODUCTION

Partially ordered sets (in short posets), lattices and frames form a constantly growing chapter which attracts the interest of many research teams (see for example [16–18, 29, 32]). Simultaneously, there is no doubt that dimensions of topological spaces are the base of many studies. There are many results for the

covering dimension, the small inductive dimension and the large inductive dimension in various classes of topological spaces (see for example [4, 11, 19, 25–27]).

Recently, the covering dimension, the small inductive dimension and the large inductive dimension had a particular interest in the class of frames (see for example [2, 7, 8, 10, 12]). In addition, several articles are devoted to the notion of dimension for finite partially ordered sets and finite lattices (see for example [1, 13–15, 22, 23, 28, 30, 31, 33–35]). The order dimension, the Krull dimension, the covering dimension and the small inductive dimension developed their own significant chapter in Dimension Lattice Theory (see for example [6, 22, 30, 35]).

In this paper, we study the notion of the large inductive dimension, Ind , in the class of finite lattices, proving properties of this dimension, like the sublattice, sum and product properties. The motivation of this study is based on the fact that posets and lattices are used as models to investigate properties (and generally topological properties) of digital images (see for example [20, 24]). Thus, the notion of dimension has its special study for finite lattices.

The paper is organized as follows. In Section 2, we present definitions that are used in the following sections. In Section 3, we present the meaning of the large inductive dimension for the class of finite lattices and some basic results. In Section 4, we compare the large inductive dimension with the small inductive dimension, proving that the large inductive dimension is always greater than or equal to the small inductive dimension. Also, we study the “gap” between these two dimensions. In addition, we present facts and various examples, investigating the relation between the large inductive dimension and the covering dimension as well as the Krull dimension and the height. Finally, in Section 5, given two finite lattices, we study the large inductive dimension of their linear sum, Cartesian product, lexicographic product and rectangular product.

2. PRELIMINARIES

In this section we recall basic definitions and notations which will be used in order to present our study. Throughout this paper we suppose that the lattices are finite. Let L be a lattice. We denote by 0_L and 1_L the bottom and the top element of L , respectively. For $x \in L$, we use the notation $\uparrow x = \{y \in L : x \leq y\}$. Since the dimension Ind is defined inductively for finite lattices (see Definition 3.1 below), we remind the reader that a *sublattice* of L is a subset of L , which is a lattice with the same meet and join operations as L . Every set $\uparrow x$ is a sublattice of L .

Moreover, we use the notation $x^* = \bigvee \{y \in L : x \wedge y = 0_L\}$. In the case where $x^* \wedge x = 0_L$, that is $x^* = \max\{y \in L : y \wedge x = 0_L\}$, the element x^* is called *pseudocomplement* of x . It is known that every finite distributive lattice is pseudocomplemented. However, in this paper we will adopt only the asterisk notation and the term “pseudocomplement” in order to have the same terminology as in

[7]. Since the lattices we deal with in this paper are not generally distributive, the use of the term “pseudocomplement” will be an agreement for the following sections, obtaining that it is an element taking as $\bigvee \{y \in L : x \wedge y = 0_L\}$.

A subset V of a lattice L is called a *cover* of L if $0_L \notin V$ and $\bigvee V = 1_L$. A cover U of a lattice L is called a *refinement* of a cover V of L , writing $U > V$, if for each $u \in U$, there exists $v \in V$ such that $u \leq v$. Especially, a cover V of L is said to be a *minimal cover* of L if $V \subseteq C$ for every refinement C of V . (We state that in related bibliography (see for example [21]) we can find variations of the meaning of minimal covers such as the minimal $\ll$ -representations of the top element of L consisting of join-irreducible elements of L . However, in our study we keep in mind the first definition of minimal covers.)

Finally, we remind the reader that a subset S of a lattice L is called *down-set* if for every $x \in L$ with $x \leq s$ for some $s \in S$, then $x \in S$. In the whole paper we denote by $\mathbb{N}$ the set of natural numbers and whenever it is necessary we write $\{1, 2, \dots\}$ for the set of natural numbers except zero.

3. THE LARGE INDUCTIVE DIMENSION AND FINITE LATTICES

In this section, we study the large inductive dimension Ind in the class of finite lattices and give basic properties of this dimension.

Definition 3.1. Let L be a finite lattice. The *large inductive dimension*, Ind , of L is defined as follows:

- (1) $\text{Ind}(L) = -1$ if and only if $L = \{0_L\}$.
- (2) $\text{Ind}(L) \leq k$, where $k \in \mathbb{N}$, if for every $a \in L$ and for every $v \in L$ such that $a \vee v = 1_L$, there exists $u \in L$ such that $u \leq v$, $a \vee u = 1_L$ and $\text{Ind}(\uparrow(u^* \vee u)) \leq k - 1$.
- (3) $\text{Ind}(L) = k$, where $k \in \mathbb{N}$, if $\text{Ind}(L) \leq k$ and $\text{Ind}(L) \not\leq k - 1$.

Analogously to Proposition 4.6 of [7], isomorphic lattices have the same large inductive dimension. Also, the simple lattice $L = \{0_L, 1_L\}$ has trivially $\text{Ind}(L) = 0$.

Example 3.2. We consider the finite lattices represented by the diagrams of Figure 1.

We have that $\text{Ind}(L_1) = 0$ and $\text{Ind}(L_2) = 1$. For example, we apply Definition 3.1 to see that $\text{Ind}(L_2) = 1$.

(1) For $a = 0_{L_2}$ and for every $v \in L_2$ with $a \vee v = 1_{L_2}$, that is $v = 1_{L_2}$, there exists $u = v = 1_{L_2}$ with $a \vee u = 1_{L_2}$ and

$$\text{Ind}(\uparrow((1_{L_2})^* \vee 1_{L_2})) = \text{Ind}(\uparrow(0_{L_2} \vee 1_{L_2})) = \text{Ind}(\uparrow 1_{L_2}) = \text{Ind}(\{1_{L_2}\}) = -1.$$

(2) For $a = y_1$ and for every $v \in L_2$ with $a \vee v = 1_{L_2}$, that is $v = 1_{L_2}$, there exists $u = v = 1_{L_2}$ with $a \vee u = 1_{L_2}$ and

$$\text{Ind}(\uparrow((1_{L_2})^* \vee 1_{L_2})) = \text{Ind}(\uparrow(0_{L_2} \vee 1_{L_2})) = \text{Ind}(\uparrow 1_{L_2}) = \text{Ind}(\{1_{L_2}\}) = -1.$$

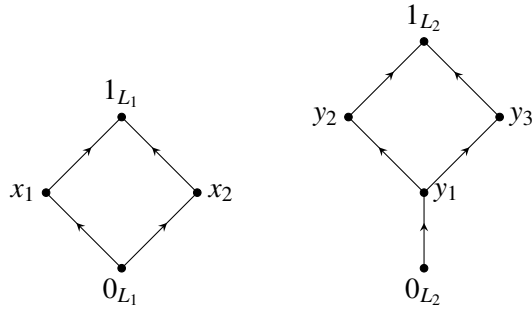


FIGURE 1. The finite lattices $(L_1, \leq_1)$ and $(L_2, \leq_2)$

(3) For $a = y_2$ and for every $v \in L_2$ with $a \vee v = 1_{L_2}$, that is $v \in \{y_3, 1_{L_2}\}$, there exists $u = y_3$ with $u \leq v$, $a \vee u = 1_{L_2}$ and

$$\text{Ind}(\uparrow(y_3^* \vee y_3)) = \text{Ind}(\uparrow(0_{L_2} \vee y_3)) = \text{Ind}(\uparrow y_3) = 0.$$

(4) For $a = y_3$ and for every $v \in L_2$ with $a \vee v = 1_{L_2}$, that is $v \in \{y_2, 1_{L_2}\}$, there exists $u = y_2$ with $u \leq v$, $a \vee u = 1_{L_2}$ and

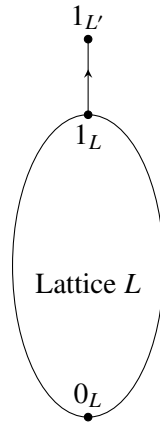
$$\text{Ind}(\uparrow(y_2^* \vee y_2)) = \text{Ind}(\uparrow(0_{L_2} \vee y_2)) = \text{Ind}(\uparrow y_2) = 0.$$

(5) For $a = 1_{L_2}$ and for every $v \in L_2$ with $a \vee v = 1_{L_2}$, that is v can be any of the elements of L_2 , there exists $u = 0_{L_2}$ with $u \leq v$, $a \vee u = 1_{L_2}$ and

$$\text{Ind}(\uparrow((0_{L_2})^* \vee 0_{L_2})) = \text{Ind}(\uparrow(1_{L_2} \vee 0_{L_2})) = \text{Ind}(\uparrow 1_{L_2}) = \text{Ind}(\{1_{L_2}\}) = -1.$$

Proposition 3.3. *Let L be a finite lattice and $k \in \mathbb{N}$ with $\text{Ind}(L) = k$. Then there exists a finite lattice L' with $\text{Ind}(L') = 0$.*

Proof. We consider the finite lattice $L' = L \cup \{1_{L'}\}$ of Figure 2, where $1_L < 1_{L'}$, that is $1_{L'}$ is the top element of L' . Then $\text{Ind}(L') = 0$.

FIGURE 2. The finite lattice L'

Indeed, for every $a \in L'$ and for every $v \in L'$ with $a \vee v = 1_{L'}$, that is $v = 1_{L'}$, there exists $u = v = 1_{L'}$ with $a \vee u = 1_{L'}$ and

$$\text{Ind}(\uparrow((1_{L'})^* \vee 1_{L'})) = \text{Ind}(\uparrow(0_{L'} \vee 1_{L'})) = \text{Ind}(\uparrow 1_{L'}) = \text{Ind}(\{1_{L'}\}) = -1.$$

□

A modification of Definition 3.1 (2) is given in the following proposition.

Theorem 3.4. *Let L be a finite lattice and $k \in \mathbb{N}$. Then $\text{Ind}(L) \leq k$ if and only if for every $a \in L$ and every $u \in \min(\{x \in L : a \vee x = 1_L\})$ we have $\text{Ind}(\uparrow(u^* \vee u)) \leq k - 1$, where $\min(\{x \in L : a \vee x = 1_L\})$ denotes the set of minimal elements of $\{x \in L : a \vee x = 1_L\}$.*

Proof. Let $\text{Ind}(L) \leq k$, $a \in L$ and $u \in \min(\{x \in L : a \vee x = 1_L\})$. We shall prove that $\text{Ind}(\uparrow(u^* \vee u)) \leq k - 1$. Clearly, $a \vee u = 1_L$. Thus, since $\text{Ind}(L) \leq k$, there exists $w \in L$ such that $w \leq u$, $w \vee a = 1_L$ and $\text{Ind}(\uparrow(w^* \vee w)) \leq k - 1$. Then $u = w$ and the result is proved.

Conversely, we shall prove that $\text{Ind}(L) \leq k$. Let $a \in L$ and $v \in L$ with $a \vee v = 1_L$. Then by assumption, every $u \in \min(\{x \in L : a \vee x = 1_L\})$ satisfying $u \leq v$ has the properties of $a \vee u = 1_L$ and $\text{Ind}(\uparrow(u^* \vee u)) \leq k - 1$. (The element u always exists since in the worst case we can take $u = v$.) Hence, $\text{Ind}(L) \leq k$. □

Theorem 3.5. *For any $k \in \{1, 2, \dots\}$, there exists a finite lattice L with $\text{Ind}(L) = k$. Also, the pseudocomplement of every element of $L \setminus \{0_L\}$ is 0_L .*

Proof. We prove the theorem by induction on k . Firstly, we shall prove that there exists a finite lattice L , in which the pseudocomplement of every element of $L \setminus \{0_L\}$ is 0_L , with $\text{Ind}(L) = 1$. Indeed, if we consider the lattice $(L_2, \leq_2)$ of Example 3.2, we have the desired result.

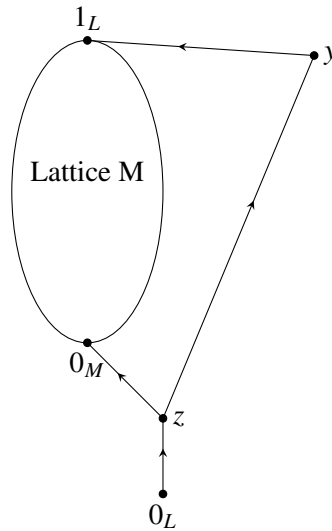


FIGURE 3. The lattice $(L, \leq)$

We suppose that there exists a finite lattice M , in which the pseudocomplement of every element of $M \setminus \{0_M\}$ is 0_M , with $\text{Ind}(M) = k - 1$, where $k > 1$. We shall prove that there exists a finite lattice L , in which the pseudocomplement of every element of $L \setminus \{0_L\}$ is 0_L , with $\text{Ind}(L) = k$.

We consider the finite lattice $L = \{0_L, z, y\} \cup M$ (see Figure 3). The partial order $\leq$ on L is defined as follows:

- (1) $z \leq w$, for every $w \in L \setminus \{0_L\}$,
- (2) $w \parallel y$, for every $w \in M$, that is, $w \not\leq y$ and $y \not\leq w$.

We observe that the pseudocomplement of every element of $L \setminus \{0_L\}$ is 0_L . We shall prove that $\text{Ind}(L) = k$. For that, we consider the following cases:

- (i) For $a = 0_L$ (respectively, $a = z$) we have $v = 1_L$ with $a \vee v = 1_L$ and thus, there exists the element $u = 1_L$ such that $u \leq v$, $a \vee u = 1_L$ and

$$\text{Ind}(\uparrow(u^* \vee u)) = -1 < k - 1.$$

(ii) For $a = 1_L$ and $v \in L$ with $a \vee v = 1_L$, there exists the element $u = 0_L$ such that $u \leq v$, $a \vee u = 1_L$ and

$$\text{Ind}(\uparrow(u^* \vee u)) = -1 < k - 1.$$

(iii) For $a = 0_M$ we have $v \in \{y, 1_L\}$ with $a \vee v = 1_L$ and thus there exists the element $u = y$ such that $u \leq v$, $a \vee u = 1_L$ and

$$\text{Ind}(\uparrow(y^* \vee y)) = \text{Ind}(\uparrow(0_L \vee y)) = \text{Ind}(\uparrow y) = 0 < k - 1.$$

(iv) Let $a = w$, where $w \in M \setminus \{0_M, 1_M\}$ (we state that $1_M = 1_L$) and $v \in M \cup \{y\}$ for which $a \vee v = 1_L$. Then we have the following cases:

(a) Let $a = w$ be any element of $M \setminus \{0_M, 1_M\}$ for which $v \in \{y, 1_L\}$. Then we consider the element $u = y$ such that

$$\text{Ind}(\uparrow(y^* \vee y)) = \text{Ind}(\uparrow(0_L \vee y)) = \text{Ind}(\uparrow y) = 0 < k - 1.$$

(b) Let $a = w$ be any element of $M \setminus \{0_M, 1_M\}$ for which $v \in M$. Since $\text{Ind}(M) = k - 1$, there exists $u \in M$ such that $u \leq v$, $a \vee u = 1_L$ and

$$\text{Ind}(\uparrow(u_M^* \vee u)) = \text{Ind}(\uparrow(0_M \vee u)) = \text{Ind}(\uparrow u) \leq k - 2,$$

and thus,

$$\text{Ind}(\uparrow(u_L^* \vee u)) = \text{Ind}(\uparrow(0_L \vee u)) = \text{Ind}(\uparrow u) \leq k - 2.$$

(v) For $a = y$ we have $v \in M$ with $a \vee v = 1_L$. Then there exists $u = 0_M$ such that $u \leq v$, $a \vee u = 1_L$ and

$$\text{Ind}(\uparrow(0_M^* \vee 0_M)) = \text{Ind}(\uparrow(0_L \vee 0_M)) = \text{Ind}(\uparrow 0_M) = \text{Ind}(M) = k - 1.$$

Therefore, $\text{Ind}(L) \leq k$. Moreover, since $\text{Ind}(L) \not\leq k - 1$, we have that $\text{Ind}(L) = k$. $\square$

We complete this section, presenting a sublattice property for the large inductive dimension. As we observe in Figure 4, a corresponding sublattice theorem does not hold without mentioning further conditions for the sublattice of a finite lattice.

Especially, for the lattice of Figure 4 we have that $\text{Ind}(L) = 0$ but for the sublattice $M = \{x_2, x_4, x_5, x_6, 1_L\}$ of L we have that $\text{Ind}(M) = 1$ and thus, $\text{Ind}(M) \not\leq \text{Ind}(L)$.

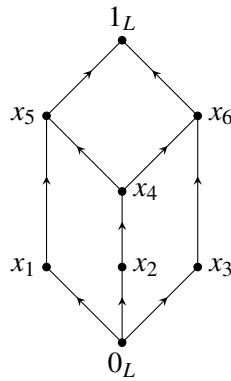


FIGURE 4. The finite lattice $(L, \leq)$

Proposition 3.6. *Let L be a finite lattice and M be a sublattice of L such that $1_L \in M$, $M \setminus \{1_L\}$ is a down-set and the lattice $\uparrow(u_M^* \vee u)$ of every element u of M is isomorphic to the lattice $\uparrow(u_L^* \vee u)$ of u in L . Then, $\text{Ind}(M) \leq \text{Ind}(L)$.*

Proof. If $\text{Ind}(L) = -1$, then, obviously, $\text{Ind}(M) = -1$. We suppose that $\text{Ind}(L) = k$, where $k \in \mathbb{N}$, and we will prove that $\text{Ind}(M) \leq k$.

Let $a \in M$ and $v \in M$ with $a \vee v = 1_M$, that is $a \vee v = 1_L$. Since $M \subseteq L$, $a, v \in L$. Also, since $\text{Ind}(L) = k$, there exists $u \in L$ such that $u \leq v$, $a \vee u = 1_L$ and $\text{Ind}(\uparrow(u_L^* \vee u)) \leq k - 1$. We consider the following cases:

- (a) If $u = 1_L$, then clearly $u \in M$ and $\text{Ind}(\uparrow(u_M^* \vee u)) = -1 = 0 - 1 \leq k - 1$.
- (b) Let $u \neq 1_L$. Since $M \setminus \{1_L\}$ is a down set and $u \leq v$, $u \in M$. By assumption since the lattices $\uparrow(u_M^* \vee u)$ and $\uparrow(u_L^* \vee u)$ are isomorphic, we have $\text{Ind}(\uparrow(u_M^* \vee u)) = \text{Ind}(\uparrow(u_L^* \vee u)) \leq k - 1$. Therefore, in each case $\text{Ind}(M) \leq k$. $\square$

4. COMPARING THE DIMENSION IND WITH OTHER LATTICE NOTIONS

In this section, we present relations between the large inductive dimension Ind and the small inductive dimension ind , the Krull dimension Kdim , the covering dimension dim and the height.

4.1. The small inductive dimension. The notion of the small inductive dimension for finite lattices is inserted in [22].

Definition 4.1. Let L be a finite lattice. The *small inductive dimension*, ind , of L is defined as follows:

- (1) $\text{ind}(L) = -1$ if and only if $L = \{0_L\}$.

- (2) $\text{ind}(L) \leq k$, where $k \in \mathbb{N}$, if for every cover V of L , there exists a cover U of L such that U is a refinement of V and $\text{ind}(\uparrow(u^* \vee u)) \leq k - 1$, for every $u \in U$.
- (3) $\text{ind}(L) = k$, where $k \in \mathbb{N}$, if $\text{ind}(L) \leq k$ and $\text{ind}(L) \not\leq k - 1$.

The following results will be very useful in our study.

Theorem 4.2 ([22]). *Let L be a finite lattice and $k \in \mathbb{N}$. Then $\text{ind}(L) \leq k$ if and only if for every minimal cover V of L we have $\text{ind}(\uparrow(v^* \vee v)) \leq k - 1$, for every $v \in V$.*

However, a modification of Theorem 4.2 is given as follows and can easily be proved following Definition 4.1 as well as Theorem 4.2.

Theorem 4.3. *Let L be a finite lattice and $k \in \mathbb{N}$. Then $\text{ind}(L) \leq k$ if and only if for every $v \in L$ such that v belongs to a minimal cover of L we have $\text{ind}(\uparrow(v^* \vee v)) \leq k - 1$.*

Theorem 4.4. *If L is a finite lattice, then $\text{ind}(L) \leq \text{Ind}(L)$.*

Proof. Clearly, if $\text{Ind}(L) = -1$, the inequality holds. We suppose that the inequality holds for all finite lattices M with $\text{Ind}(M) \leq k - 1$. Let L be a finite lattice such that $\text{Ind}(L) = k$, $w \in L$ and $W = \{w_1, \dots, w_m\}$ be a minimal cover of L such that $w \in W$. Then $w = w_i$ for some $i = 1, \dots, m$. By Theorem 4.3 it suffices to prove that $\text{ind}(\uparrow(w_i \vee w_i^*)) \leq k - 1$.

Let $a = \bigvee_{j=1, j \neq i}^m w_j$ and $v = w_i$ with $a \vee w_i = 1_L$. Since $\text{Ind}(L) = k$, there exists $u \in L$ such that $u \leq w_i$, $a \vee u = 1_L$ and $\text{Ind}(\uparrow(u \vee u^*)) \leq k - 1$ and by inductive assumption, $\text{ind}(\uparrow(u \vee u^*)) \leq k - 1$. Then the set $U = \{w_1, \dots, u, \dots, w_m\}$, where U is the set W replacing w_i by u , is a cover of L which refines W . Since W is minimal, $W \subseteq U$. Thus, $w_i = u$ and then $\text{ind}(\uparrow(w_i \vee w_i^*)) \leq k - 1$. Thus, $\text{ind}(L) \leq k$. $\square$

Remark 4.5. Generally, the small inductive dimension and the large inductive dimension are different notions of dimensions for finite lattices. For example, for the lattice of Figure 5 we have $\text{Ind}(L) = 1$ and $\text{ind}(L) = 0$.

Especially, for the large inductive dimension the element which gives us that $\text{Ind}(L) = 1$ is the element $a = x_7$ for which $v \in \{x_6, x_8, 1_L\}$ with $a \vee v = 1_L$ and for $u = x_6$ we have that $\text{Ind}(\uparrow(x_6 \vee x_6^*)) = 0$. For the small inductive dimension we observe that for all minimal covers V of L we have $\text{ind}(\uparrow(v \vee v^*)) = -1$, for every $v \in V$. Thus, by Theorem 4.2 we have $\text{ind}(L) = 0$.

We state that the family

$$\mathcal{MC} = \{\{x_7, x_8\}, \{x_2, x_3, x_7\}, \{x_2, x_8\}, \{x_3, x_8\}\}$$

consists of all the minimal covers of L . The fact that gives us the difference between the two dimensions ind and Ind is the study of the element x_6 . For this element we have $\text{Ind}(\uparrow(x_6 \vee x_6^*)) = 0$ and therefore,

$\text{Ind}(L) = 1$. Moreover, we can see that $\text{ind}(\uparrow(x_6 \vee x_6^*)) = 0$. Thus, if the element x_6 belongs to a minimal cover of L , then $\text{ind}(L) = 1$ and hence, $\text{ind}(L) = \text{Ind}(L)$. However, we observe that there is no minimal cover V of L with $x_6 \in V$ and thus, x_6 does not make the small inductive dimension equal to 1. All minimal covers of L and their elements give us that $\text{ind}(L) = 0$.

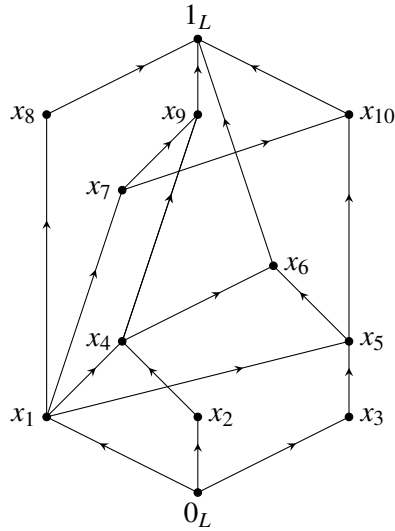


FIGURE 5. The finite lattice $(L, \leq)$

We can generalize Remark 4.5 in order to construct a finite lattice L with arbitrary large inductive dimension and $\text{Ind}(L) = \text{ind}(L) + 1$.

Proposition 4.6. *Let $k \in \{1, 2, \dots\}$. Then there exists a finite lattice M such that $\text{ind}(M) = k - 1$ and $\text{Ind}(M) = k$.*

Proof. We show the proposition by induction on k . Remark 4.5 proves the proposition for the case $k = 1$. We consider the lattice L of Figure 5. On this lattice we replace the sublattice $(\{x_6, 1_L\}, \leq)$ with a sublattice N such that:

- (1) $\text{Ind}(N) = k - 1$, where $k \in \{2, 3, \dots\}$,
- (2) $x_6 = 0_N$,
- (3) $x_6 \leq w \leq 1_L$, for every $w \in N$,
- (4) the pseudocomplement of every element of $N \setminus \{0_N\}$ is 0_N and
- (5) each $u \in L \setminus N \cup \{z \in L : z \leq x_6\}$ and each element $v \in N \setminus \{0_N, 1_N\}$ are incomparable (see Figure 6).

We state that the existence of the lattice N follows by Theorem 3.5. We write this new lattice as M . Also, for the elements $a = x_7$ and $v = x_6$, we have $x_7 \vee x_6 = 1_M$ and

$$\{u \in L : u \leq x_6 \text{ and } x_7 \vee u = 1_M\} = \{x_6\}.$$

Since

$$\text{Ind}(\uparrow(x_6 \vee x_6^*)) = \text{Ind}(\uparrow(x_6 \vee 0_M)) = \text{Ind}(\uparrow x_6) = \text{Ind}(N) = k - 1,$$

we have $\text{Ind}(M) \geq k$. Furthermore, following Definition 3.1 we can see that $\text{Ind}(M) \leq k$.

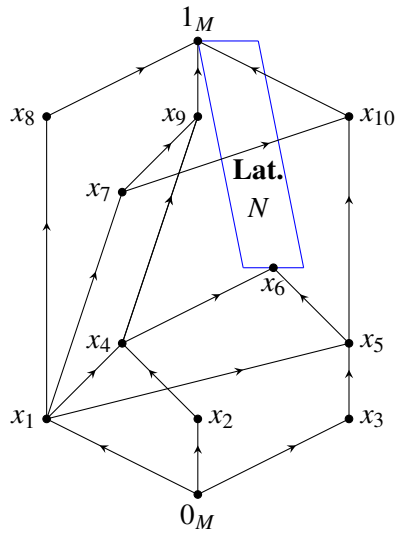


FIGURE 6. The finite lattice $(M, \leq)$

Moreover, the construction of the lattice M leads us to observe that the minimal covers of M are the minimal covers of the family $\mathcal{MC}$ of Remark 4.5, and the minimal covers of N . For each minimal cover $V \in \mathcal{MC}$ we have $\text{ind}(\uparrow(v \vee v^*)) = -1$, for every $v \in V$. Also, since the construction of N follows the construction given in [22, Theorem 4], we have $\text{ind}(N) = k - 1$. Let U be a minimal cover of N . Since $\text{ind}(N) \leq k - 1$, by Theorem 4.2 we have

$$\text{ind}(\uparrow(u \vee u_N^*)) = \text{ind}(\uparrow(u \vee x_6)) = \text{ind}(\uparrow u) \leq k - 2,$$

for every $u \in U$, where u_N^* denotes the pseudocomplement of u in the lattice N . Thus,

$$\text{ind}(\uparrow(u \vee u_M^*)) = \text{ind}(\uparrow(u \vee 0_M)) = \text{ind}(\uparrow u) \leq k - 2,$$

for every $u \in U$, where u_M^* denotes the pseudocomplement of u in the lattice M . Therefore, by Theorem 4.2, we have that $\text{ind}(M) \leq k - 1$. We prove that $\text{ind}(M) = k - 1$. Since $\text{ind}(N) = k - 1$, there exist a minimal cover W of N and $w \in W$ such that

$$\text{ind}(\uparrow(w \vee w_N^*)) = \text{ind}(\uparrow(w \vee x_6)) = \text{ind}(\uparrow w) = k - 2.$$

Hence, $\text{ind}(\uparrow(w \vee w_M^*)) = \text{ind}(\uparrow(w \vee 0_M)) = \text{ind}(\uparrow w) = k - 2$. By Theorem 4.2 we conclude that $\text{ind}(M) = k - 1$. $\square$

Remark 4.7. We observe that the “gap” between the dimensions ind and Ind can be arbitrarily large. For that, we consider the lattice $(Q, \leq)$ of Figure 7. For this lattice we have that $\text{Ind}(Q) = 3$ and $\text{ind}(Q) = 0$.

Firstly, we describe the construction of the lattice $(Q, \leq)$. We consider the lattice M of Figure 6, where $N = \{1_M, x_6\} \cup \{w_1, \dots, w_6\}$ is the sublattice given in Theorem 3.5 for $k = 2$, that is $\text{Ind}(N) = 2$. On this lattice M , we add the following elements:

(I) for each $w \in N \setminus \{1_M, x_6\}$ we assign an element x_w such that:

- (1) $0_M < x_w < w$,
- (2) for every $w_1, w_2 \in N \setminus \{1_M, x_6\}$ with $w_1 \neq w_2$, $x_{w_1} \neq x_{w_2}$,
- (3) x_w is incomparable with any other element of M ,

(II) an element s such that:

- (1) $0_M < s < x_w$, for every x_w ,

(III) elements t, r, u as in Figure 7 in order to succeed that the constructed poset is a lattice.

We write this new lattice as Q . Using the same argument as in Proposition 4.6 we have $\text{Ind}(Q) = 3$, considering the elements $a = x_7$ and $v = x_6$. Also, the minimal covers of Q are the family

$$\mathcal{MC}' = \{\{s, x_8\}, \{x_2, x_3, x_7\}, \{x_2, x_8\}, \{x_3, x_8\}\}$$

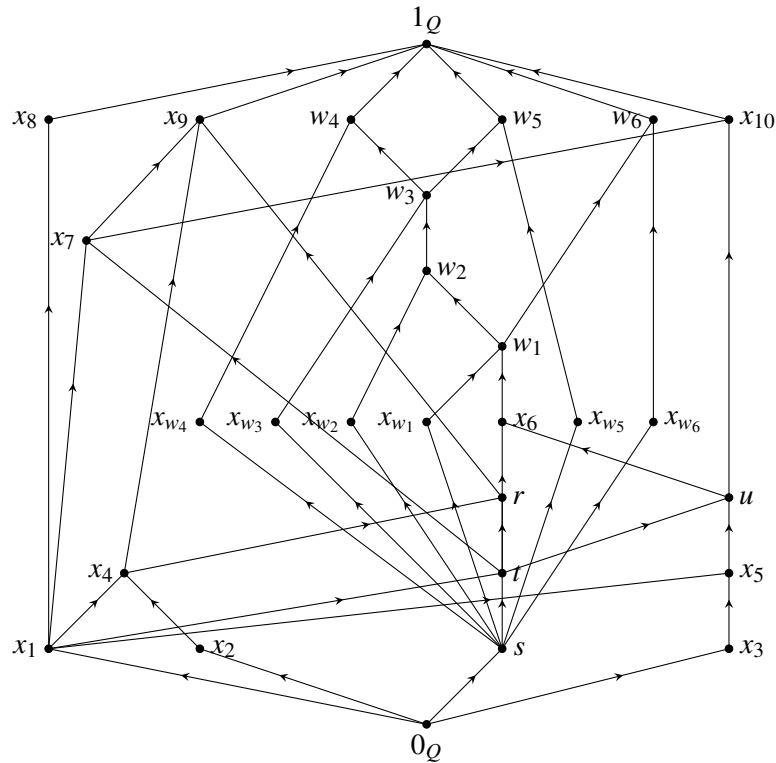
and the minimal covers consisting of elements of the set $\{x_{w_1}, \dots, x_{w_6}\}$. For each minimal cover $V \in \mathcal{MC}'$ we have $\text{ind}(\uparrow(v \vee v^*)) = -1$, for every $v \in V$. Also, for each $x_w \in Q$ we have

$$\text{ind}(\uparrow(x_w \vee x_w^*)) = \text{ind}(\uparrow(x_w \vee 1_Q)) = \text{ind}(\uparrow 1_Q) = \text{ind}(\{1_Q\}) = -1.$$

Therefore, $\text{ind}(Q) = 0$.

Now, we can generalize Remark 4.7 in order to construct a finite lattice P for which $\text{Ind}(P) - \text{ind}(P)$ is arbitrarily large.

Proposition 4.8. *Let $k \in \{1, 2, \dots\}$. Then there exists a finite lattice P such that $\text{ind}(P) = 0$ and $\text{Ind}(P) = k$.*

FIGURE 7. The finite lattice $(Q, \leq)$

Proof. We shall prove the proposition by induction on k . If $k = 1$, then it follows by Proposition 4.6 that we get the desired result. Suppose $k \geq 2$ and we have a finite lattice N such that $\text{ind}(N) = 0$ and $\text{Ind}(N) = k - 1$. We consider the lattice M of Figure 6. On this lattice M , we add the following elements as in Figure 8:

(I) for each $w \in N \setminus \{1_M, x_6\}$ we assign an element x_w such that:

- (1) $0_M < x_w < w$,
- (2) for every $w_1, w_2 \in N \setminus \{1_M, x_6\}$ with $w_1 \neq w_2$, $x_{w_1} \neq x_{w_2}$,
- (3) x_w is incomparable with any other element of M ,

(II) an element s such that:

- (1) $0_M < s < x_w$, for every x_w ,

(III) elements t, r, u as in Figure 8 in order to succeed that the constructed poset is a lattice.

We write this new lattice as P . Using the same argument as in Proposition 4.6 we have $\text{Ind}(P) = k$. Also, the minimal covers of P are the family MC' of Remark 4.7 and the minimal covers consisting of

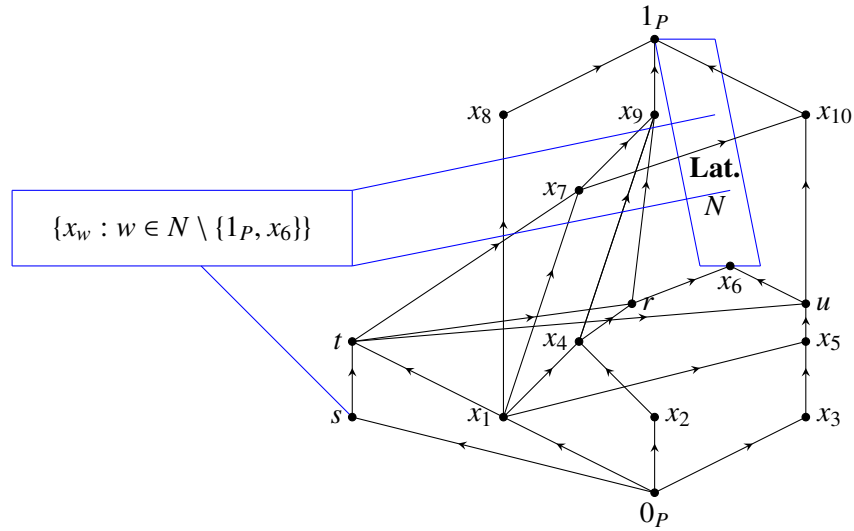


FIGURE 8. The finite lattice $(P, \leq)$

elements of the set $\{x_w : w \in N \setminus \{1_P, x_6\}\}$. For each minimal cover $V \in \mathcal{MC}'$ we have $\text{ind}(\uparrow(v \vee v^*)) = -1$, for every $v \in V$. Also, for each $x_w \in P$ we have

$$\text{ind}(\uparrow(x_w \vee x_w^*)) = \text{ind}(\uparrow(x_w \vee 1_P)) = \text{ind}(\uparrow 1_P) = \text{ind}(\{1_P\}) = -1.$$

Therefore, $\text{ind}(P) = 0$. □

However, we observe that if we add some conditions for finite lattices, then we can have the equality of these two dimensions.

Lemma 4.9. *Let L be a finite lattice and $M = \uparrow x$, where $x \in L$. If for every minimal cover V of L we have $|V| \leq 2$, then for every minimal cover W of M we have also $|W| \leq 2$.*

Proof. Let W be a minimal cover of M . We shall prove that $|W| \leq 2$. We have that W is a cover of L .

(1) If W is a minimal cover of L , then by assumption $|W| \leq 2$.

(2) We suppose that W is not a minimal cover of L . Then we consider the following cases:

(a) If $1_L \in W$ that is $1_M \in W$ (as $1_L = 1_M$), then $W = \{1_M\}$ is the only minimal cover of M for which $|W| = 1$.

(b) We state that $1_L \notin W$. By [14, Lemma 3.7], there exists a minimal cover R of L which refines W and $1_L \notin R$. By assumption we have that $|R| = \{c, d\}$. Then $c \leq a$ and $d \leq b$, for some $a, b \in W$. Moreover, $a \vee b = 1_L = 1_M$. Indeed, if $a \vee b = r$, where $r \neq 1_L$, then the set $\{c, d\}$ is not a cover of L ,

which is a contradiction. Thus, $\{a, b\}$ is a cover of M , which refines W . Since W is a minimal cover of M and $a, b \in W$, we have that $W = \{a, b\}$, that is $|W| = 2$.

Therefore, in each case we have that $|W| \leq 2$. $\square$

Proposition 4.10. *Let L be a finite lattice. If for every minimal cover V of L we have $|V| \leq 2$, then $\text{ind}(L) = \text{Ind}(L)$.*

Proof. By Theorem 4.4 we have that $\text{ind}(L) \leq \text{Ind}(L)$. Thus, it suffices to prove that $\text{Ind}(L) \leq \text{ind}(L)$. Clearly, if $\text{ind}(L) = -1$, the inequality holds. We suppose that the proposition holds for all finite lattices M with $\text{ind}(M) \leq k - 1$. Let L be a finite lattice such that $\text{ind}(L) = k$. We shall prove that $\text{Ind}(L) \leq k$.

(1) For $a = 1_L$ and any $v \in L$ with $a \vee v = 1_L$, there exists the element $u = 0_L$ such that $u \leq v$, $a \vee u = 1_L$ and $\text{Ind}(\uparrow(u \vee u^*)) = -1 \leq k - 1$.

(2) For $a = 0_L$ and $v = 1_L$ with $a \vee v = 1_L$, there exists the element $u = 1_L$ such that $u \leq v$, $a \vee u = 1_L$ and $\text{Ind}(\uparrow(u \vee u^*)) = -1 \leq k - 1$.

(3) Let $a \in L \setminus \{0_L, 1_L\}$ and $v \in L$ with $a \vee v = 1_L$. If $v = 1_L$, we have the trivial fact $\text{Ind}(\uparrow(v \vee v^*)) = -1$. Therefore, we suppose that $v \neq 1_L$ and we study the following cases.

(a) If $\{a, v\}$ is a minimal cover of L , then since $\text{ind}(L) = k$, by Theorem 4.2 we have $\text{ind}(\uparrow(v \vee v^*)) \leq k - 1$. Moreover, by Lemma 4.9, the assumption of the proposition also holds for the lattice $\uparrow(v \vee v^*)$. Therefore, by induction, we have that $\text{Ind}(\uparrow(v \vee v^*)) \leq k - 1$. Thus, in Definition 3.1, if $u = v$, then u is an element of L for which $a \vee u = 1_L$ and $\text{Ind}(\uparrow(u \vee u^*)) \leq k - 1$.

(b) We suppose that $\{a, v\}$ is not a minimal cover of L . We state that $1_L \notin \{a, v\}$. By [14, Lemma 3.7], there exists a minimal cover R of L which refines $\{a, v\}$ and $1_L \notin R$. By assumption we have that $R = \{c, d\}$. Then $c \leq a$ and $d \leq v$. Also, since $\text{ind}(L) = k$, by Theorem 4.2 we have $\text{ind}(\uparrow(d \vee d^*)) \leq k - 1$ and by induction (as Lemma 4.9 shows that the assumption of the proposition also holds for the lattice $\uparrow(d \vee d^*)$), we have that $\text{Ind}(\uparrow(d \vee d^*)) \leq k - 1$. Moreover, we observe that $a \vee d = 1_L$. Indeed, if $a \vee d = r$, where $r \neq 1_L$, then the set $\{c, d\}$ is not a cover of L , which is a contradiction. Thus, in Definition 3.1, if $u = d$, then u is an element of L for which $u \leq v$, $a \vee u = 1_L$ and $\text{Ind}(\uparrow(u \vee u^*)) \leq k - 1$.

Therefore, in each case we have that $\text{Ind}(L) \leq k$. $\square$

4.2. The Krull dimension. A non-empty subset F of a lattice $(L, \leq)$ is called *filter* if F has the following properties:

- (1) $F \neq \emptyset$,
- (2) If $x \in F$ and $x \leq y$, then $y \in F$,
- (3) If $x, y \in F$, then $x \wedge y \in F$.

A filter F is called *prime* if for every $x, y \in L$ with $x \vee y \in F$, we have $x \in F$ or $y \in F$. The set of all prime filters of a lattice L is usually denoted by $\mathcal{PF}(L)$.

Definition 4.11 ([31]). If $\mathcal{PF}(L) \neq \emptyset$, then the *Krull dimension* of L is defined as follows:

$$\text{Kdim}(L) = \sup\{k : \text{there exist prime filters } F_0 \subset F_1 \subset \cdots \subset F_k\}.$$

Also, in the study [15], the Krull dimension is studied through join prime elements and matrices. An element $x \in L$ is said to be *join prime* if it is non-zero and the inequality $x \leq a \vee b$ implies $x \leq a$ or $x \leq b$, for all $a, b \in L$.

Example 4.12. We show by examples that the dimensions Ind and Kdim for finite lattices are in general different.

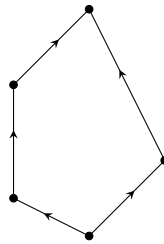


FIGURE 9. The pentagon

- (1) The large inductive dimension of the pentagon (see Figure 9) is equal to 0, but since it is not a Boolean Algebra, its Krull dimension is not equal to 0 (see [31]).
- (2) We consider the finite lattice $(L, \leq)$, represented by the diagram of Figure 10.

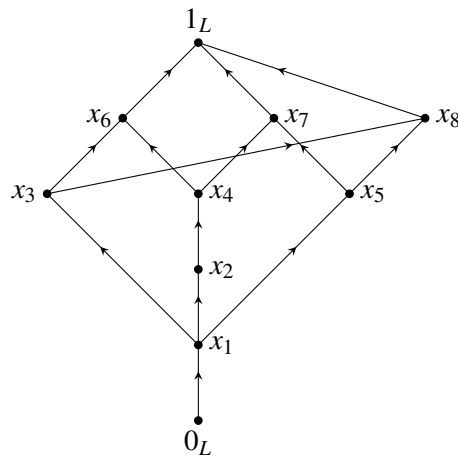


FIGURE 10. The finite lattice $(L, \leq)$

We have that $\text{Ind}(L) = 2$ and $\text{Kdim}(L) = 1$. Indeed, the elements which give us that $\text{Ind}(L) = 2$ are $a = x_8$ and $v = x_2$ for which $a \vee v = 1_L$. Then

$$\{u \in L : u \leq x_2 \text{ and } a \vee u = 1_L\} = \{x_2\}$$

and $\text{Ind}(\uparrow(x_2 \vee x_2^*)) = \text{Ind}(\uparrow x_2) = 1$. Hence, $\text{Ind}(L) \geq 2$. It is easy to show that $\text{Ind}(L) \leq 2$. Moreover, the join prime elements of L are the elements x_1, x_2, x_3 and x_5 and thus, by Proposition 4.5 (5) of [15] we have that $\text{Kdim}(L) = 1$.

4.3. The covering dimension. Let L be a finite lattice. The *order* of a subset C of L , denoted by $\text{ord}(C)$, is defined to be k , where $k \in \mathbb{N}$ if and only if the infimum of any $k + 2$ distinct elements of C is 0_L and there exist $k + 1$ distinct elements of C whose infimum is not 0_L .

Definition 4.13 ([14]). The function dim , called *covering dimension*, with domain the class of all finite lattices and range the set $\mathbb{N}$ is defined as follows:

- (1) $\text{dim}(L) \leq k$, where $k \in \mathbb{N}$, if and only if for every cover C of L , there exists a cover R of L , refinement of C with $\text{ord}(R) \leq k$.
- (2) $\text{dim}(L) = k$, where $k \in \mathbb{N}$, if $\text{dim}(L) \leq k$ and $\text{dim}(L) \not\leq k - 1$.

In Theorem 3.8 of [14], it was proved that $\text{dim}(L) \leq k$, where $k \in \mathbb{N}$, if and only if for every minimal cover V of L we have $\text{ord}(V) \leq k$. That is,

$$\text{dim}(L) = \max\{\text{ord}(V) : V \text{ is minimal cover of } L\}$$

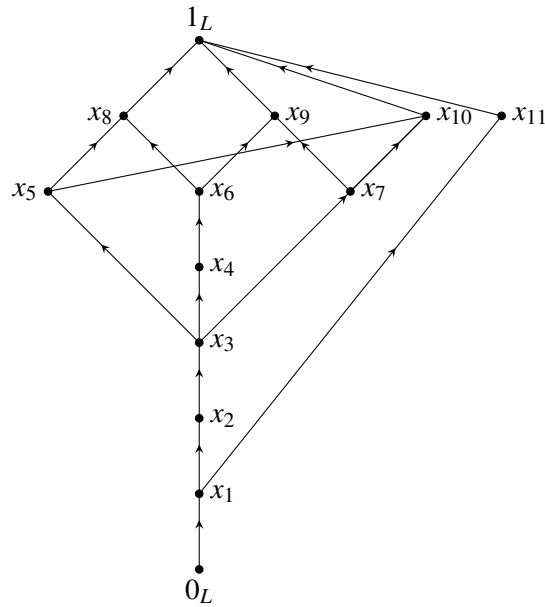
(see Corollary 3.9 of [14]).

Remark 4.14. We show by examples that the dimensions Ind and dim for finite lattices are in general different.

(1) We consider the lattice of Figure 11. We have that $\text{Ind}(L) = 3$ and $\text{dim}(L) = 2$. Indeed, the elements which give us that $\text{Ind}(L) = 3$ are $a = x_{11}$ and $v = x_2$ for which $a \vee v = 1_L$. Then

$$\{u \in L : u \leq x_2 \text{ and } a \vee u = 1_L\} = \{x_2\}$$

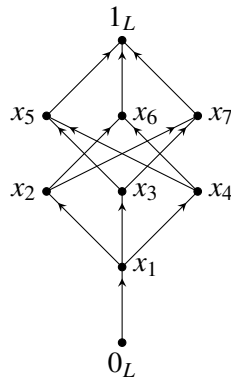
and $\text{Ind}(\uparrow(x_2 \vee x_2^*)) = \text{Ind}(\uparrow x_2) = 2$. Hence, $\text{Ind}(L) \geq 3$. It is easy to show that $\text{Ind}(L) \leq 3$. Moreover, for all minimal covers V of L , that is $V \in \{\{x_2, x_{11}\}, \{x_4, x_5, x_7\}\}$, we have $\text{ord}(V) \leq 2$. Thus, $\text{dim}(L) = 2$.

FIGURE 11. The finite lattice $(L, \leq)$

(2) For the lattice L of Figure 12 we have $\dim(L) = 2$ and $\text{Ind}(L) = 1$. Indeed, let $a \in L \setminus \{0_L, 1_L\}$ and $v \in L$ with $a \vee v = 1_L$. It is clear that $a \vee x_1 = a \neq 1_L$. Thus, for each $u \in L$ with $u \leq v$ and $a \vee u = 1_L$, we have $u \neq x_1$. Further, for each $i = 2, \dots, 7$ $x_i^* = 0_L$ and hence $\uparrow(x_i \vee x_i^*) = \uparrow x_i$. It is easy to show that $\text{Ind}(\uparrow x_i) = 0$. Thus, we have $\text{Ind}(L) \geq 1$. Further, we notice that $x_2 \vee x_5 = 1_L$,

$$\{u \in L : u \leq x_5 \text{ and } x_2 \vee u = 1_L\} = \{x_5\}$$

and $\text{Ind}(\uparrow x_5) = 0$. Hence $\text{Ind}(L) = 1$. For the covering dimension, the only minimal cover of L is the set $V = \{x_2, x_3, x_4\}$ for which $\text{ord}(V) = 2$. Thus, $\dim(L) = 2$.

FIGURE 12. The finite lattice $(L, \leq)$

4.4. The height.

Definition 4.15 ([5]). The *height* of a finite lattice L , denoted by $h(L)$, is defined as follows:

$$h(L) = \max\{k : \text{there exist } a_0, \dots, a_k \in L \text{ such that } 0_L = a_0 < \dots < a_k = 1_L\}.$$

Example 4.16. For the lattice L_1 of Figure 1 we have that $\text{Ind}(L_1) = 0$ and $h(L_1) = 2$ and for the totally ordered set $L = \{x_1, \dots, x_n, 0_L, 1_L\}$ we have that $h(L) = n + 1$ and $\text{Ind}(L) = 0$.

Proposition 4.17. For every finite lattice L , $\text{Ind}(L) < h(L)$.

Proof. We prove the inequality by induction on $h(L)$. Clearly if $h(L) = 0$, then $L = \{1_L\}$ and thus, $\text{Ind}(L) = -1$, proving that the relation of the proposition holds. We suppose that the inequality holds for all finite lattices M with $h(M) < k$, for $k > 1$, and we shall prove it for k . Let L be a finite lattice with $h(L) = k$, $v \in L$ such that v belongs to a minimal cover V of L . By Theorem 4.3 it suffices to prove that $\text{Ind}(\uparrow(v^* \vee v)) \leq k - 2$. Since $\uparrow(v^* \vee v) \neq L$, we have that $h(\uparrow(v^* \vee v)) < h(L)$ and thus, $h(\uparrow(v^* \vee v)) \leq k - 1$. By inductive hypothesis we have that $\text{Ind}(\uparrow(v^* \vee v)) < k - 1$, proving that $\text{Ind}(L) \leq k - 1$ or equivalently $\text{Ind}(L) < k$. $\square$

5. SUM AND PRODUCT PROPERTIES FOR IND

In this section we study properties of the large inductive dimension. Especially, we study the large inductive dimension of the linear sum and kinds of products of finite lattices.

Definition 5.1 ([13]). The *linear sum* of two posets $(L_1, \leq_1)$ and $(L_2, \leq_2)$ such that $L_1 \cap L_2 = \emptyset$, denoted by $L_1 \oplus L_2$, is the poset $(L_1 \cup L_2, \leq)$, where the relation $\leq$ is defined as follows:

$$x \leq y \Leftrightarrow \begin{cases} x, y \in L_1 \text{ and } x \leq_1 y \\ x, y \in L_2 \text{ and } x \leq_2 y \\ x \in L_1, y \in L_2. \end{cases}$$

Clearly, if $(L_1, \leq_1)$ and $(L_2, \leq_2)$ are lattices, then $(L_1 \oplus L_2, \leq)$ is also a lattice.

Remark 5.2. (1) The lattices $(L_1, \leq_1)$ and $(L_3, \leq_3)$ of Figures 1 and 13, respectively, verifies the claim $\text{Ind}(L_1 \oplus L_3) \neq \text{Ind}(L_3 \oplus L_1)$. We have $\text{Ind}(L_1) = \text{Ind}(L_3) = 0$, $\text{Ind}(L_1 \oplus L_3) = 0$ and $\text{Ind}(L_3 \oplus L_1) = 1$.



FIGURE 13. The finite lattice $(L_3, \leq_3)$

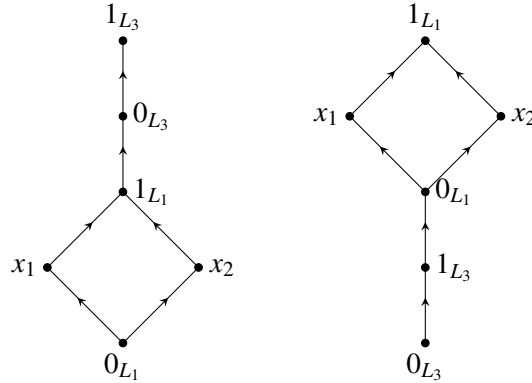


FIGURE 14. The finite lattices $(L_1 \oplus L_3, \leq)$ and $(L_3 \oplus L_1, \leq^*)$

(2) The lattice $(L_3 \oplus L_1, \leq^*)$ of Figure 14 also verifies that the assertion

$$\text{Ind}(L_3 \oplus L_1) \leq \text{Ind}(L_3) + \text{Ind}(L_1)$$

does not hold for all finite lattices L_3 and L_1 .

Proposition 5.3. *Let $(L_1, \leq_1)$ and $(L_2, \leq_2)$ be two finite lattices such that the pseudocomplement of every element of $L_2 \setminus \{0_{L_2}\}$ is 0_{L_2} . Then*

$$\text{Ind}(L_2) \leq \text{Ind}(L_1 \oplus L_2).$$

Proof. Let $\text{Ind}(L_1 \oplus L_2) = k$, where $k \in \mathbb{N}$, $a \in L_2$ and $v \in L_2$ such that $a \vee v = 1_{L_2}$.

(1) If $a = 1_{L_2}$ and $v \in L_2$, then there exists the element $u = 0_{L_2}$ such that $u \leq_2 v$ and $\text{Ind}(\uparrow(u \vee u_{L_2}^*)) = -1 \leq k - 1$, where $u_{L_2}^*$ denotes the pseudocomplement of u in the lattice L_2 .

(2) If $a = 0_{L_2}$, then the only element $v \in L_2$ for which $a \vee v = 1_{L_2}$ is the element 1_{L_2} . Thus, for this element $u = 1_{L_2}$ we have $\text{Ind}(\uparrow(u \vee u_{L_2}^*)) = -1 \leq k - 1$.

(3) Let $a \in L_2 \setminus \{1_{L_2}, 0_{L_2}\}$. Then $a, v \in L_1 \oplus L_2$ with $a \vee v = 1_{L_1 \oplus L_2}$. Since $\text{Ind}(L_1 \oplus L_2) = k$, there exists $u \in L_1 \oplus L_2$ such that $u \leq v$, $a \vee u = 1_{L_1 \oplus L_2}$ (and therefore, $a \vee u = 1_{L_2}$) and $\text{Ind}(\uparrow(u \vee u_{L_1 \oplus L_2}^*)) \leq k - 1$, where $u_{L_1 \oplus L_2}^*$ denotes the pseudocomplement of u in the lattice $L_1 \oplus L_2$.

By the definition of the linear sum $L_1 \oplus L_2$, in order to have $a \vee u = 1_{L_2}$, $u \in L_2$ and thus,

$$\begin{aligned} u_{L_1 \oplus L_2}^* &= \bigvee \{x \in L_1 \oplus L_2 : x \wedge u = 0_{L_1 \oplus L_2}\} \\ &= \bigvee \{x \in L_1 \oplus L_2 : x \wedge u = 0_{L_1}\} \\ &= 0_{L_1}. \end{aligned}$$

Therefore,

$$\text{Ind}(\uparrow(u \vee u_{L_1 \oplus L_2}^*)) = \text{Ind}(\uparrow(u \vee 0_{L_1})) = \text{Ind}(\uparrow u) \leq k - 1$$

and thus,

$$\text{Ind}(\uparrow(u \vee u_{L_2}^*)) = \text{Ind}(\uparrow(u \vee 0_{L_2})) = \text{Ind}(\uparrow u) \leq k - 1.$$

Hence, $\text{Ind}(L_2) \leq k$. □

Definition 5.4 ([13]). If $(L_1, \leq_1)$ and $(L_2, \leq_2)$ are two posets, then their *Cartesian product* is the poset $(L_1 \times L_2, \leq)$, where the relation $\leq$ is defined as follows: $(x_1, y_1) \leq (x_2, y_2) \Leftrightarrow x_1 \leq_1 x_2$ and $y_1 \leq_2 y_2$.

If $(L_1, \leq_1)$ and $(L_2, \leq_2)$ are finite lattices, then $(L_1 \times L_2, \leq)$ is also a finite lattice. Also, if $(x_1, y_1), (x_2, y_2) \in L_1 \times L_2$, then:

- (1) $(x_1, y_1) \vee (x_2, y_2) = (x_1 \vee x_2, y_1 \vee y_2)$ and
- (2) $(x_1, y_1) \wedge (x_2, y_2) = (x_1 \wedge x_2, y_1 \wedge y_2)$.

Lemma 5.5 ([9, 22]). *If L_1 and L_2 are finite lattices and $(x, y) \in L_1 \times L_2$, then the following are satisfied:*

- (1) $\uparrow(x, y) = \uparrow x \times \uparrow y$,
- (2) $(x, y)^* = (x^*, y^*)$.

Theorem 5.6. *Let $(L_1, \leq_1)$ and $(L_2, \leq_2)$ be finite lattices. Then,*

$$\text{Ind}(L_1 \times L_2) = \max\{\text{Ind}(L_1), \text{Ind}(L_2)\}.$$

Proof. In order to prove the inequality

$$\text{Ind}(L_1 \times L_2) \leq \max\{\text{Ind}(L_1), \text{Ind}(L_2)\},$$

we shall apply induction with respect to the number

$$k(L_1, L_2) = \max\{\text{Ind}(L_1), \text{Ind}(L_2)\}.$$

If $k(L_1, L_2) = -1$, then our inequality holds. Moreover, if $\text{Ind}(L_1) = -1$ and $\text{Ind}(L_2) \geq 0$ (respectively, $\text{Ind}(L_2) = -1$ and $\text{Ind}(L_1) \geq 0$), then the lattices $L_1 \times L_2$ and L_2 (respectively, $L_1 \times L_2$ and L_1) are isomorphic and therefore, $\text{Ind}(L_1 \times L_2) = \text{Ind}(L_2)$ (respectively, $\text{Ind}(L_1 \times L_2) = \text{Ind}(L_1)$).

We assume that the inequality holds for every pair of finite lattices L_1 and L_2 with $k(L_1, L_2) < k$, where $k \geq 0$, and we consider finite lattices L_1 and L_2 such that $\text{Ind}(L_1) = n \geq 0$, $\text{Ind}(L_2) = m \geq 0$ and $\max\{n, m\} = k$. We shall prove that $\text{Ind}(L_1 \times L_2) \leq k$.

Let $(a, b) \in L_1 \times L_2$ and $(u, v) \in L_1 \times L_2$ such that $(a, b) \vee (u, v) = (1_{L_1}, 1_{L_2})$. Then $a, u \in L_1$ with $a \vee u = 1_{L_1}$ and $b, v \in L_2$ with $b \vee v = 1_{L_2}$. Since $\text{Ind}(L_1) = n$, there exists $x \in L_1$ with $x \leq_1 u$, $x \vee a = 1_{L_1}$ and

$$\text{Ind}(\uparrow(x \vee x^*)) \leq n - 1.$$

Similarly, since $\text{Ind}(L_2) = m$, there exists $y \in L_2$ with $y \leq_2 v$, $y \vee b = 1_{L_2}$ and

$$\text{Ind}(\uparrow(y \vee y^*)) \leq m - 1.$$

We consider the element $(x, y) \in L_1 \times L_2$. Then $(x, y) \leq (u, v)$ and $(a, b) \vee (x, y) = (1_{L_1}, 1_{L_2})$. We shall prove that

$$\text{Ind}(\uparrow((x, y)^* \vee (x, y))) \leq \max\{n, m\} - 1.$$

Considering Lemma 5.5 we have that

$$\begin{aligned} \text{Ind}(\uparrow((x, y)^* \vee (x, y))) &= \text{Ind}(\uparrow((x^*, y^*) \vee (x, y))) \\ &= \text{Ind}(\uparrow(x^* \vee x, y^* \vee y)) \\ &= \text{Ind}(\uparrow(x^* \vee x) \times \uparrow(y^* \vee y)). \end{aligned}$$

By inductive hypothesis, we have that

$$\begin{aligned} \text{Ind}(\uparrow(x^* \vee x) \times \uparrow(y^* \vee y)) &\leq \max\{\text{Ind}(\uparrow(x^* \vee x)), \text{Ind}(\uparrow(y^* \vee y))\} \\ &\leq \max\{n-1, m-1\} \\ &= \max\{n, m\} - 1. \end{aligned}$$

Therefore, $\text{Ind}(L_1 \times L_2) \leq \max\{n, m\}$.

We shall prove the inequality

$$\max\{\text{Ind}(L_1), \text{Ind}(L_2)\} \leq \text{Ind}(L_1 \times L_2).$$

If $\text{Ind}(L_1 \times L_2) = -1$, then $L_1 \times L_2 = \{1_{L_1 \times L_2}\}$ and thus, $L_1 = \{1_{L_1}\}$ and $L_2 = \{1_{L_2}\}$ and the inequality holds.

We suppose that the inequality holds for every pair of finite lattices L_1 and L_2 with $\text{Ind}(L_1 \times L_2) < k$ and we shall prove it for k . Let $\text{Ind}(L_1 \times L_2) = k$. We shall prove that $\text{Ind}(L_1) \leq k$ and $\text{Ind}(L_2) \leq k$.

Let $a \in L_1$ and $u \in L_1$ with $a \vee u = 1_{L_1}$. Let also $b \in L_2$ and $v \in L_2$ with $b \vee v = 1_{L_2}$. Then $(a, b), (u, v) \in L_1 \times L_2$ with $(a, b) \vee (u, v) = (1_{L_1}, 1_{L_2})$. Since $\text{Ind}(L_1 \times L_2) = k$, there exists $(x, y) \in L_1 \times L_2$ such that $(x, y) \leq (u, v)$, $(a, b) \vee (x, y) = (1_{L_1}, 1_{L_2})$ and

$$\text{Ind}(\uparrow((x, y)^* \vee (x, y))) \leq k - 1.$$

That is, $x \in L_1$ with $x \leq_1 u$ and $a \vee x = 1_{L_1}$ and $y \in L_2$ with $y \leq_2 v$ and $b \vee y = 1_{L_2}$. It suffices to prove that $\text{Ind}(\uparrow(x^* \vee x)) \leq k - 1$ and $\text{Ind}(\uparrow(y^* \vee y)) \leq k - 1$.

Considering Lemma 5.5 we have that

$$\begin{aligned} \text{Ind}(\uparrow((x, y)^* \vee (x, y))) &= \text{Ind}(\uparrow((x^*, y^*) \vee (x, y))) \\ &= \text{Ind}(\uparrow(x^* \vee x, y^* \vee y)) \\ &= \text{Ind}(\uparrow(x^* \vee x) \times \uparrow(y^* \vee y)). \end{aligned}$$

By inductive hypothesis,

$$\max\{\text{Ind}(\uparrow(x^* \vee x)), \text{Ind}(\uparrow(y^* \vee y))\} \leq \text{Ind}(\uparrow(x^* \vee x) \times \uparrow(y^* \vee y)) \leq k - 1.$$

Therefore, $\text{Ind}(\uparrow(x^* \vee x)) \leq k - 1$ and $\text{Ind}(\uparrow(y^* \vee y)) \leq k - 1$. Thus, $\text{Ind}(L_1) \leq k$ and $\text{Ind}(L_2) \leq k$. $\square$

Thus, in contrast to Remark 5.2 for the linear sum, Theorem 5.6 tends to the following corollaries which show the respect of Ind to a kind of Cartesian product theorem and the property of commutativity.

Corollary 5.7. *Let $(L_1, \leq_1)$ and $(L_2, \leq_2)$ be finite lattices. Then,*

$$(1) \text{Ind}(L_1 \times L_2) \leq \text{Ind}(L_1) + \text{Ind}(L_2),$$

$$(2) \text{Ind}(L_1 \times L_2) = \text{Ind}(L_2 \times L_1).$$

Definition 5.8 ([13]). For two lattices $(L_1, \leq_1)$ and $(L_2, \leq_2)$ the *lexicographic product* $L_1 \diamond L_2$ is the lattice $(L_1 \times L_2, \leq)$, where the relation $\leq$ is defined as follows:

$$(x_1, y_1) \leq (x_2, y_2) \Leftrightarrow \begin{cases} x_1 <_1 x_2 \text{ or} \\ x_1 = x_2 \text{ and } y_1 \leq_2 y_2. \end{cases}$$

By Definition 4.8 of [14] for every $(x, y), (x', y') \in L_1 \diamond L_2$ we have that

$$(x, y) \wedge (x', y') = \begin{cases} (x, y), & \text{if } x < x' \\ (x', y'), & \text{if } x' < x \\ (x, y \wedge y'), & \text{if } x = x' \\ (x \wedge x', 1_{L_2}), & \text{if } x \parallel x' \end{cases}$$

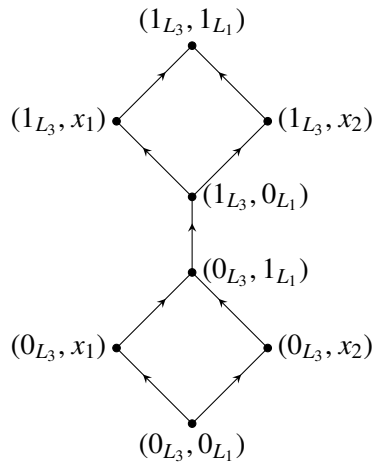
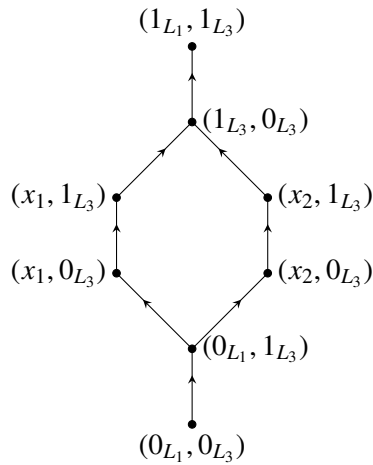
and

$$(x, y) \vee (x', y') = \begin{cases} (x, y), & \text{if } x' < x \\ (x', y'), & \text{if } x < x' \\ (x, y \vee y'), & \text{if } x = x' \\ (x \vee x', 0_{L_2}), & \text{if } x \parallel x'. \end{cases}$$

Remark 5.9. (1) The lattices $(L_1, \leq_1)$ and $(L_3, \leq_3)$ of Figures 1 and 13, respectively, verifies that the inequality

$$\text{Ind}(L_3 \diamond L_1) \leq \text{Ind}(L_3) + \text{Ind}(L_1)$$

does not hold for all finite lattices L_1 and L_3 . We have that $\text{Ind}(L_1) = \text{Ind}(L_3) = 0$. For the finite lattice $(L_3 \diamond L_1, \leq)$, which is represented by the diagram of Figure 15, we have that $\text{Ind}(L_3 \diamond L_1) = 1$.


 FIGURE 15. The finite lattice $(L_3 \diamond L_1, \leq)$

 FIGURE 16. The finite lattice $(L_1 \diamond L_3, \leq)$

(2) The lattices $(L_1, \leq_1)$ and $(L_3, \leq_3)$ of Figures 1 and 13, respectively, verifies that the inequality

$$\text{Ind}(L_3 \diamond L_1) = \text{Ind}(L_1 \diamond L_3)$$

does not hold for all finite lattices L_1 and L_3 . As we have seen in Figure 15, $\text{Ind}(L_3 \diamond L_1) = 1$. However, $\text{Ind}(L_1 \diamond L_3) = 0$. Indeed, the diagram of $L_1 \diamond L_3$ is given in Figure 16, from which we can easily see that $\text{Ind}(L_1 \diamond L_3) = 0$.

Proposition 5.10. *Let $(L_1, \leq_1)$ and $(L_2, \leq_2)$ be two finite lattices such that the pseudocomplement of every element of $L_2 \setminus \{0_{L_2}\}$ is 0_{L_2} . Then*

$$\text{Ind}(L_2) \leq \text{Ind}(L_1 \diamond L_2).$$

Proof. Let $\text{Ind}(L_1 \diamond L_2) = k, k \in \mathbb{N}, a \in L_2$ and $v \in L_2$ with $a \vee v = 1_{L_2}$.

(1) If $a = 1_{L_2}$, then for every $v \in L_2$ we have $a \vee v = 1_{L_2}$ and thus, there exists the element $u' = 0_{L_2}$ such that $u' \leq_2 v, a \vee u' = 1_{L_2}$ and

$$\text{Ind}(\uparrow(u' \vee (u')_{L_2}^*)) = -1 \leq k - 1,$$

where $(u')_{L_2}^*$ denotes the pseudocomplement of u' in L_2 .

(2) If $a = 0_{L_2}$, then for the only element $v = 1_{L_2}$ we have $a \vee v = 1_{L_2}$. Thus, there exists the element $u' = 1_{L_2}$ such that $u' \leq_2 v, a \vee u' = 1_{L_2}$ and

$$\text{Ind}(\uparrow(u' \vee (u')_{L_2}^*)) = -1 \leq k - 1.$$

(3) Let $a \in L_2 \setminus \{0_{L_2}, 1_{L_2}\}$. Then $(1_{L_1}, a)$ and $(1_{L_1}, v)$ belong to $L_1 \diamond L_2$ and $(1_{L_1}, a) \vee (1_{L_1}, v) = (1_{L_1}, 1_{L_2})$. Since $\text{Ind}(L_1 \diamond L_2) = k$, there exists $(u, u') \in L_1 \diamond L_2$ such that $(u, u') \leq (1_{L_1}, v), (u, u') \vee (1_{L_1}, a) = (1_{L_1}, 1_{L_2})$ and

$$\text{Ind}(\uparrow((u, u') \vee (u, u')_{L_1 \diamond L_2}^*)) \leq k - 1,$$

where $(u, u')_{L_1 \diamond L_2}^*$ denotes the pseudocomplement of (u, u') in $L_1 \diamond L_2$. Since $(u, u') \leq (1_{L_1}, v), a \neq 1_{L_2}$ and $(u, u') \vee (1_{L_1}, a) = (1_{L_1}, 1_{L_2})$, we have that $u = 1_{L_1}, u' \leq_2 v$ and $u' \vee a = 1_{L_2}$. Since $(u')_{L_2}^* = 0_{L_2}$ by the assumption of the proposition, we have $(1_{L_1}, u')_{L_1 \diamond L_2}^* = (0_{L_1}, 0_{L_2})$ and thus,

$$\text{Ind}(\uparrow(u, u')) = \text{Ind}(\uparrow((u, u') \vee (u, u')_{L_1 \diamond L_2}^*)) \leq k - 1.$$

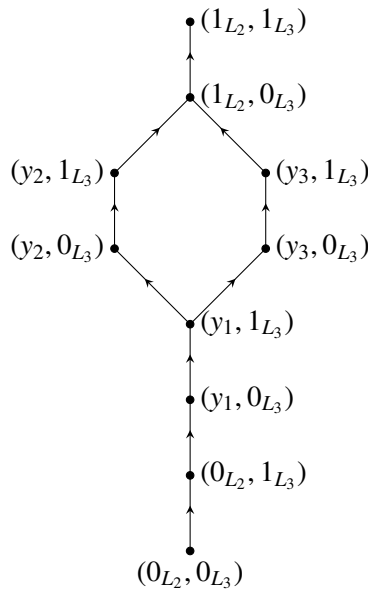
Since $\uparrow(1_{L_1}, u') = \uparrow 1_{L_1} \times \uparrow u'$, we have that

$$\text{Ind}(\uparrow u') = \text{Ind}(\{1_{L_1}\} \times \uparrow u') = \text{Ind}(\uparrow(1_{L_1}, u')) \leq k - 1$$

and thus,

$$\text{Ind}(\uparrow(u' \vee (u')_{L_2}^*)) = \text{Ind}(\uparrow(u' \vee 0_{L_2})) = \text{Ind}(\uparrow u') \leq k - 1.$$

Therefore, $\text{Ind}(L_2) \leq k$. □

FIGURE 17. The finite lattice $(L_2 \diamond L_3, \leqslant)$

Remark 5.11. The dimension Ind of the Cartesian product $L_1 \times L_2$ of two finite lattices L_1 and L_2 differs from the dimension Ind of their lexicographic product.

(1) For the lattices $(L_1, \leqslant_1)$ and $(L_3, \leqslant_3)$ of Figures 1 and 13, respectively, we have $\text{Ind}(L_1) = \text{Ind}(L_3) = 0$ and by Theorem 5.6, $\text{Ind}(L_3 \times L_1) = 0$. For their lexicographic product (see Figure 15) we have $\text{Ind}(L_3 \diamond L_1) = 1$.

(2) For the lattices $(L_2, \leqslant_2)$ and $(L_3, \leqslant_3)$ of Figures 1 and 13, respectively, we have $\text{Ind}(L_2) = 1$, $\text{Ind}(L_3) = 0$ and by Theorem 5.6, $\text{Ind}(L_2 \times L_3) = 1$. For their lexicographic product (see Figure 17) we have $\text{Ind}(L_2 \diamond L_3) = 0$.

Definition 5.12 ([1, 3]). The *rectangular product* of two lattices $(L_1, \leqslant_1)$ and $(L_2, \leqslant_2)$ is the lattice $(L_1 \square L_2, \leqslant)$, where

$$L_1 \square L_2 = \{(x, y) \in L_1 \times L_2 : x \neq 0_{L_1} \text{ and } y \neq 0_{L_2}\} \cup \{(0_{L_1}, 0_{L_2})\}$$

and the relation $\leqslant$ is defined as follows:

$$(x_1, y_1) \leqslant (x_2, y_2) \Leftrightarrow x_1 \leqslant_1 x_2 \text{ and } y_1 \leqslant_2 y_2.$$

That is, $L_1 \square L_2$ is the subset of $L_1 \times L_2$ obtained by removing the points of the form $(x, 0_{L_2})$ and $(0_{L_1}, y)$ with $x \neq 0_{L_1}$ and $y \neq 0_{L_2}$. Also, if $(x_1, y_1), (x_2, y_2) \in L_1 \square L_2$, then:

- (1) $(x_1, y_1) \vee (x_2, y_2) = (x_1 \vee x_2, y_1 \vee y_2)$ and
- (2) $(x_1, y_1) \wedge (x_2, y_2) = \begin{cases} (x_1 \wedge x_2, y_1 \wedge y_2), & \text{if } x_1 \wedge x_2 \neq 0_{L_1} \text{ and } y_1 \wedge y_2 \neq 0_{L_2} \\ (0_{L_1}, 0_{L_2}), & \text{otherwise.} \end{cases}$

In order to study the relation of the rectangular product of two finite lattices with their large inductive dimension, we recall the following fact.

Proposition 5.13 ([22]). *Let L_1 and L_2 be two finite lattices and $(a, b) \in L_1 \square L_2$. Then*

$$(a, b)^* = \begin{cases} (a^*, b^*) = (0_{L_1}, 0_{L_2}), & \text{if } a^* = 0_{L_1} \text{ and } b^* = 0_{L_2} \\ (1_{L_1}, 1_{L_2}), & \text{if } a^* \neq 0_{L_1} \text{ and } b^* \neq 0_{L_2} \\ (1_{L_1}, b^*), & \text{if } a^* = 0_{L_1} \text{ and } b^* \neq 0_{L_2} \\ (a^*, 1_{L_2}), & \text{if } a^* \neq 0_{L_1} \text{ and } b^* = 0_{L_2}. \end{cases}$$

Up to now, the above discussions related the dimension Ind and the properties of linear sum, Cartesian product and lexicographic product tend to study the behavior of this dimension without mentioning additional lattice properties, verifying that Ind respects such properties. However, this seems to fail for the rectangular product.

Let us observe that in Figure 4 we have seen a finite lattice L for which $\text{Ind}(\uparrow x_2) \not\leq \text{Ind}(L)$. This fact leads us to insert a new lattice property as follows. This property will be useful in order to succeed a kind of rectangular product theorem given in Theorem 5.15.

Definition 5.14. Let L be a finite lattice. If the sublattice property of the form $\text{Ind}(\uparrow x) \leq \text{Ind}(L)$ holds for every $x \in L$, then we say that L has the *SP-property*.

Theorem 5.15. *Let $(L_1, \leq_1)$ and $(L_2, \leq_2)$ be two finite lattices such that the SP-property holds for all $\uparrow \kappa$ and $\uparrow \lambda$, where $\kappa \in L_1$ and $\lambda \in L_2$. Then*

$$\text{Ind}(L_1 \square L_2) \leq \max\{\text{Ind}(L_1), \text{Ind}(L_2)\} + 1.$$

Proof. Let $\text{Ind}(L_1) = n$ and $\text{Ind}(L_2) = m$, where $n, m \in \mathbb{N}$. We shall prove that $\text{Ind}(L_1 \square L_2) \leq \max\{n, m\} + 1$. 1. Let $(a, b) \in L_1 \square L_2$. Then we study the following cases:

- (1) If $(a, b) = (0_{L_1}, 0_{L_2})$, then for every $(u, v) \in L_1 \square L_2$ with $(a, b) \vee (u, v) = (1_{L_1}, 1_{L_2})$ (that is, $(u, v) = (1_{L_1}, 1_{L_2})$), there is $(x, y) = (1_{L_1}, 1_{L_2}) \in L_1 \square L_2$ such that $(x, y) \leq (u, v)$, $(a, b) \vee (x, y) = (1_{L_1}, 1_{L_2})$ and

$$\text{Ind}(\uparrow((x, y) \vee (x, y)^*)) = -1 < \max\{n, m\}.$$

- (2) Let $a \neq 0_{L_1}$, $b \neq 0_{L_2}$.

(2a) If $(a, b) = (1_{L_1}, 1_{L_2})$ and $(u, v) \in L_1 \square L_2$ such that $(a, b) \vee (u, v) = (1_{L_1}, 1_{L_2})$, then there exists $(x, y) = (0_{L_1}, 0_{L_2}) \in L_1 \square L_2$ such that $(x, y) \leq (u, v)$, $(a, b) \vee (x, y) = (1_{L_1}, 1_{L_2})$ and

$$\text{Ind}(\uparrow((x, y) \vee (x, y)^*)) = -1 < \max\{n, m\}.$$

(2b) Let $(a, b) \in (L_1 \square L_2) \setminus \{(0_{L_1}, 0_{L_2}), (1_{L_1}, 1_{L_2})\}$ and $(u, v) \in L_1 \square L_2$ such that $(a, b) \vee (u, v) = (1_{L_1}, 1_{L_2})$. We shall prove that there exists $(x, y) \in L_1 \square L_2$ such that $(x, y) \leq (u, v)$, $(a, b) \vee (x, y) = (1_{L_1}, 1_{L_2})$ and

$$\text{Ind}(\uparrow((x, y) \vee (x, y)^*)) \leq \max\{n, m\}.$$

We consider as (x, y) the element (u, v) . Then by Proposition 5.13 we have the following cases:

(A) $(u, v) \vee (u, v)^* = (u, v) \vee (u^*, 1_{L_2}) = (u \vee u^*, 1_{L_2})$ and thus, by Lemma 5.5 and Theorem 5.6, we have that

$$\text{Ind}(\uparrow((u, v) \vee (u, v)^*)) = \max\{\text{Ind}(\uparrow(u \vee u^*)), -1\}.$$

By SP-property for the lattice L_1 we also have that $\text{Ind}(\uparrow(u \vee u^*)) \leq \text{Ind}(L_1)$ and thus,

$$\text{Ind}(\uparrow((u, v) \vee (u, v)^*)) \leq \max\{n, -1\}.$$

(B) $(u, v) \vee (u, v)^* = (u, v) \vee (1_{L_1}, v^*) = (1_{L_1}, v \vee v^*)$ and thus, this case is similar to Case (A).

(C) $(u, v) \vee (u, v)^* = (u, v) \vee (0_{L_1}, 0_{L_2}) = (u, v)$ and thus, by Lemma 5.5 and Theorem 5.6, we have that

$$\text{Ind}(\uparrow((u, v) \vee (u, v)^*)) = \max\{\text{Ind}(\uparrow u), \text{Ind}(\uparrow v)\}.$$

By SP-property for the lattices L_1 and L_2 we also have $\text{Ind}(\uparrow u) \leq \text{Ind}(L_1)$ and $\text{Ind}(\uparrow v) \leq \text{Ind}(L_2)$. Thus,

$$\text{Ind}(\uparrow((u, v) \vee (u, v)^*)) \leq \max\{n, m\}.$$

(D) $(u, v) \vee (u, v)^* = (u, v) \vee (1_{L_1}, 1_{L_2}) = (1_{L_1}, 1_{L_2})$ and thus,

$$\text{Ind}(\uparrow((u, v) \vee (u, v)^*)) = -1 < \max\{n, m\}.$$

Therefore, in each case we have that

$$\text{Ind}(\uparrow((u, v) \vee (u, v)^*)) \leq \max\{n, m\}.$$

Thus,

$$\text{Ind}(L_1 \square L_2) \leq \max\{n, m\} + 1.$$

□

Corollary 5.16. Let $(L_1, \leq_1)$ and $(L_2, \leq_2)$ be two finite lattices such that the SP-property holds for all $\uparrow\kappa$ and $\uparrow\lambda$, where $\kappa \in L_1$ and $\lambda \in L_2$. Then

$$\text{Ind}(L_1 \square L_2) \leq \text{Ind}(L_1) + \text{Ind}(L_2) + 1.$$

Theorem 5.17. *Let $(L_1, \leq_1)$ and $(L_2, \leq_2)$ be two finite lattices such that the pseudocomplement of every $w \in L_i \setminus \{0_{L_i}\}$ is 0_{L_i} , for $i = 1, 2$. Then*

$$\text{Ind}(L_1 \times L_2) \leq \text{Ind}(L_1 \sqcup L_2).$$

Proof. By Theorem 5.6, it suffices to prove that

$$\max\{\text{Ind}(L_1), \text{Ind}(L_2)\} \leq \text{Ind}(L_1 \sqcup L_2).$$

Let $\text{Ind}(L_1 \sqcup L_2) = k$, where $k \in \mathbb{N}$. Then we shall prove that $\text{Ind}(L_i) \leq k$, for $i = 1, 2$. Firstly, we shall prove that $\text{Ind}(L_1) \leq k$. Let $a \in L_1$ and $v \in L_1$ with $a \vee v = 1_{L_1}$.

(1) If $a = 1_{L_1}$ and $v \in L_1$, we consider on the element $u = 0_{L_1}$. Then, $u \leq_1 v$, $a \vee u = 1_{L_1}$ and $\text{Ind}(\uparrow(u \vee u^*)) = \text{Ind}(\{1_{L_1}\}) = -1 \leq k - 1$.

(2) If $a = 0_{L_1}$, then the only element $v \in L_1$ for which $a \vee v = 1_{L_1}$ is the element 1_{L_1} . Thus, for this element $u = 1_{L_1}$ we have $\text{Ind}(\uparrow(u \vee u^*)) = -1 \leq k - 1$.

(3) Let $a \in L_1 \setminus \{0_{L_1}, 1_{L_1}\}$ and $v \in L_1$ with $a \vee v = 1_{L_1}$. Then $v \neq 0_{L_1}$ and thus, $(a, 1_{L_2}), (v, 1_{L_2}) \in L_1 \sqcup L_2$ with $(a, 1_{L_2}) \vee (v, 1_{L_2}) = (1_{L_1}, 1_{L_2})$. Since $\text{Ind}(L_1 \sqcup L_2) = k$, there exists $(x, y) \in L_1 \sqcup L_2$ such that $(x, y) \leq (v, 1_{L_2})$, $(x, y) \vee (a, 1_{L_2}) = (1_{L_1}, 1_{L_2})$ and

$$\text{Ind}(\uparrow((x, y) \vee (x, y)^*)) \leq k - 1.$$

Then $x \leq_1 v$ and $x \vee a = 1_{L_1}$. Also, by assumption we have

$$\text{Ind}(\uparrow(x \vee x^*)) = \text{Ind}(\uparrow(x \vee 0_{L_1})) = \text{Ind}(\uparrow x).$$

Thus, it suffices to prove that $\text{Ind}(\uparrow x) \leq k - 1$. By Proposition 5.13 and our assumption we have that

$$(x, y) \vee (x, y)^* = (x, y) \vee (x^*, y^*) = (x, y) \vee (0_{L_1}, 0_{L_2})$$

and thus, by Lemma 5.5 and Theorem 5.6,

$$\max\{\text{Ind}(\uparrow x), \text{Ind}(\uparrow y)\} = \text{Ind}(\uparrow(x, y)) = \text{Ind}(\uparrow((x, y) \vee (x, y)^*))$$

and therefore,

$$\max\{\text{Ind}(\uparrow x), \text{Ind}(\uparrow y)\} \leq k - 1,$$

that is $\text{Ind}(\uparrow x) \leq k - 1$. Therefore, $\text{Ind}(L_1) \leq k$.

Similarly, we can prove that $\text{Ind}(L_2) \leq k$. Hence, $\max\{\text{Ind}(L_1), \text{Ind}(L_2)\} \leq k$ and thus, by Theorem 5.6 we have that $\text{Ind}(L_1 \times L_2) \leq k$. □

Corollary 5.18. *Let $(L_1, \leq_1)$ and $(L_2, \leq_2)$ be two finite lattices such that:*

- (1) *the SP-property holds for all $\uparrow\kappa$ and $\uparrow\lambda$, where $\kappa \in L_1$ and $\lambda \in L_2$ and*
- (2) *the pseudocomplement of every $w \in L_i \setminus \{0_{L_i}\}$ is 0_{L_i} , for $i = 1, 2$.*

Then

$$\text{Ind}(L_1 \times L_2) \leq \text{Ind}(L_1 \sqcup L_2) \leq \text{Ind}(L_1 \times L_2) + 1.$$

Proof. By Theorems 5.6 and 5.15, we have $\text{Ind}(L_1 \sqcup L_2) \leq \text{Ind}(L_1 \times L_2) + 1$. Moreover, by Theorem 5.17 we have $\text{Ind}(L_1 \times L_2) \leq \text{Ind}(L_1 \sqcup L_2)$. $\square$

Remark 5.19. The dimension Ind of the Cartesian product $L_1 \times L_2$ of two finite lattices L_1 and L_2 differs from the dimension Ind of their rectangular product.

(1) We consider the finite lattices, which are represented in Figures 18 and 19. We have that $\text{Ind}(L_1) = \text{Ind}(L_2) = 0$, $\text{Ind}(L_1 \sqcup L_2) = 1$ and by Theorem 5.6, $\text{Ind}(L_1 \times L_2) = 0$.



FIGURE 18. The finite lattices $(L_1, \leq_1)$ and $(L_2, \leq_2)$

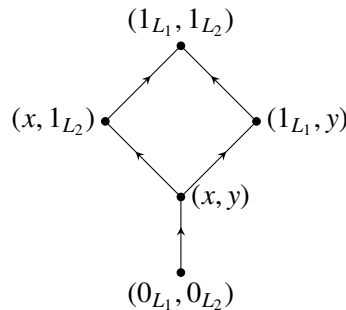


FIGURE 19. The finite lattice $(L_1 \times L_2, \leq)$

(2) We consider the finite lattices of Figure 1. Then $\text{Ind}(L_1) = 0$ and $\text{Ind}(L_2) = 1$ and thus, by Theorem 5.6 we have that $\text{Ind}(L_2 \times L_1) = 1$. For the rectangular product $L_2 \sqcup L_1$, which is represented by the diagram of Figure 20, we have $\text{Ind}(L_2 \sqcup L_1) = 0$.

Especially, for $\text{Ind}(L_2 \sqcup L_1) = 0$, we follow the argument (2) of Definition 3.1. We observe that for every $(a, b) \in L_2 \sqcup L_1$ and for every $(u, v) \in L_2 \sqcup L_1$ with $(a, b) \vee (u, v) = (1_{L_2}, 1_{L_1})$ we always find an

element $(x, y) \in L_2 \square L_1$ such that $(x, y) \leq (u, v)$, $(a, b) \vee (u, v) = (1_{L_2}, 1_{L_1})$ and

$$\text{Ind}(\uparrow((x, y)^* \vee (x, y))) = \text{Ind}(\{(1_{L_2}, 1_{L_1})\}) = -1.$$

Thus, $\text{Ind}(L_2 \square L_1) = 0$.

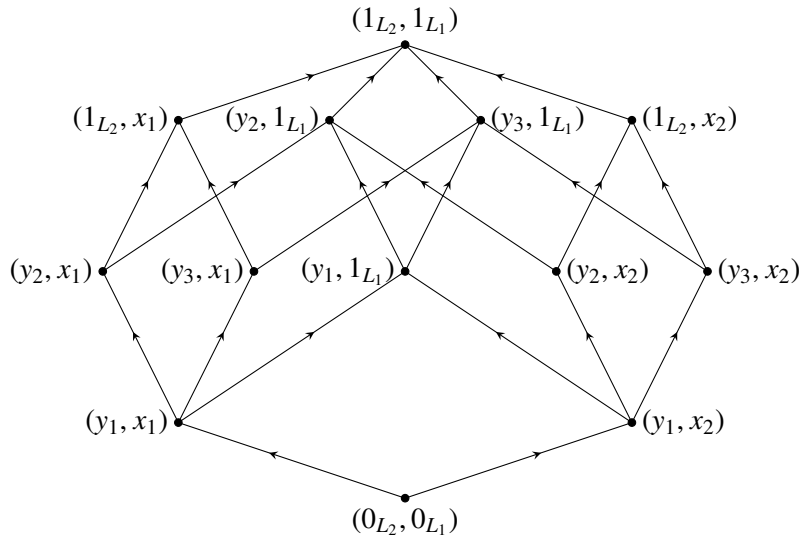


FIGURE 20. The finite lattice $(L_2 \square L_1, \leq)$

Remark 5.20. The dimension Ind of the lexicographic product $L_1 \diamond L_2$ of two finite lattices L_1 and L_2 differs from the dimension Ind of their rectangular product.

(1) We consider the finite lattices $(L_1, \leq_1)$ and $(L_3, \leq_3)$ of Figures 1 and 13, respectively. Then as we have seen in Figure 15, we have $\text{Ind}(L_3 \diamond L_1) = 1$. For their rectangular product, the figure of which is isomorphic to the lattice $(L_1, \leq_1)$ of Figure 1, we have $\text{Ind}(L_3 \square L_1) = 0$.

(2) We consider the lattices $(L_2, \leq_2)$ and $(L_3, \leq_3)$ of Figures 1 and 13, respectively. As we have seen in Figure 17, we have $\text{Ind}(L_2 \diamond L_3) = 0$. For their rectangular product, the figure of which is isomorphic to the lattice $(L_2, \leq_2)$ of Figure 1, we have $\text{Ind}(L_2 \square L_3) = 1$.

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